EPA Data Project Phase 2

Due 11/29

Data Analysis Visualization Impact Assessment Prediction and Sensitivity Analysis

Phase 1 establishes the database foundation for the EPA Data application through data download, file upload, validation, database storage, search, filtering, sorting, pagination, record details, and filtered export. Phase 2 will extend that foundation into a data analysis and decision-support system. Users will be able to summarize data, compare facilities and units, create interactive visualizations, estimate potential environmental and health impacts with EPA TRACI factors, develop prediction models, and examine how changes in selected inputs affect model outputs.

1 Project Goals

Phase 2 will enable users to:

1. Explore EPA power-sector data using descriptive statistics.
2. Compare facilities, generating units, fuels, states, and reporting periods.
3. Visualize emissions and operational measurements interactively.
4. Estimate selected potential environmental and health impacts using EPA TRACI factors.
5. Identify trends, relationships, and unusual observations.
6. Build and evaluate prediction models.
7. Perform sensitivity and scenario analysis.
8. Save or export analytical results.
9. Explain analytical methods, assumptions, results, and limitations clearly.

2 Intended Users

- Environmental researchers
- Power-sector analysts
- Government and regulatory personnel
- Facility managers
- Students studying databases, data science, or environmental analytics
- Members of the public interested in power-sector emissions

3 Data Requirements

Phase 2 will use two complementary EPA data sources:

- CAMPD emissions and operating data for multiple reporting years, including CO2, SO2, NOx, gross load, heat input, and facility and unit attributes
- EPA TRACI 2.2 characterization factors for converting selected air emissions into common potential-impact indicators

CAMPD will supply the measured or reported pollutant quantities and electricity-generation data. TRACI will supply reference factors for impact categories such as global warming, acidification, smog formation, and selected human-health effects. The first implementation should use only factors that can be matched clearly to CAMPD CO2, SO2, or NOx emissions to air.

The system should store the TRACI version, pollutant name, impact category, factor value, and factor unit so that every calculated result can be reproduced and interpreted correctly.

4 Proposed Functions

4.1 Analytical Dataset Builder

Users should be able to define the records included in an analysis by selecting:

- Reporting-year range
- State, facility, or generating unit
- Primary fuel and unit type
- Control technology or emissions program
- Minimum operating time or gross load

The application should display the number of selected records, facilities, units, and years; a missing-value summary; and the variables available for analysis. Filters should be reproducible through URL parameters or a saved analysis configuration.

4.2 Descriptive Statistics

For a selected numerical variable, the system should calculate:

- Count and missing-value count
- Minimum and maximum
- Mean and median
- Standard deviation
- First and third quartiles
- Interquartile range and selected percentiles

Grouped statistics should support questions such as average CO2 emissions by state, median NOx intensity by fuel type, total SO2 emissions by year, and average heat rate by unit type.

4.3 Derived Performance Indicators

Phase 2 should calculate normalized indicators in addition to raw totals. Examples include:

- CO2 intensity = CO2 mass divided by gross load
- SO2 intensity = SO2 mass divided by gross load
- NOx intensity = NOx mass divided by gross load
- Heat rate = heat input divided by gross load
- Year-over-year absolute and percentage change
- Operating-utilization measures when the required fields are available

The application must handle division by zero and missing measurements correctly. Undefined values should remain missing rather than being replaced automatically with zero.

4.4 Interactive Visualization Dashboard

The dashboard should support:

- Line charts for emissions, gross load, and intensity over time
- Bar charts comparing states, fuels, facilities, units, or control technologies
- Histograms and box plots showing distributions

- Scatter plots and correlation heat maps showing relationships
- Rankings of high- and low-emitting facilities or units
- State-level maps or facility-location maps when geographic data are available

Visualizations should include descriptive titles, measurement units, legends, tooltips, filters, and data or image download. Plotly.js is recommended because it supports interactive charts, zooming, tooltips, and image export.

4.5 Comparative Analysis

Users should be able to compare:

- Facilities and units
- States, fuels, and unit types
- Units with and without a selected pollution control
- Different reporting periods

Comparisons should include summary statistics, absolute and percentage differences, rankings, charts, and normalized indicators. Optional statistical methods may include confidence intervals, correlation, two-sample tests, analysis of variance, and nonparametric comparisons. Assumptions and conclusions must be explained in plain language.

4.6 Environmental and Health Impact Indicators

The application should provide a simple TRACI impact-assessment function. It will convert selected CAMPD CO2, SO2, and NOx emissions from short tons to kilograms, match each pollutant with an applicable TRACI air-emission characterization factor, and multiply the pollutant mass by that factor.

How the two datasets connect

CAMPD provides the inventory data: the amount of each pollutant emitted by a facility or generating unit during a selected period. TRACI provides the characterization factor that represents the potential contribution of one kilogram of a substance to a specific impact category. The application connects the datasets by matching the CAMPD pollutant to the corresponding TRACI substance and the air-emission medium.

- CAMPD CO2 mass maps to the verified TRACI carbon-dioxide air-emission factor.
- CAMPD SO2 mass maps to the verified TRACI sulfur-dioxide air-emission factor.
- CAMPD NOx mass maps only to a verified TRACI nitrogen-oxides air-emission factor; students should not rely on approximate name matching.

Calculation procedure

- Filter and aggregate the CAMPD records at the chosen facility, unit, state, fuel, or year level.
- Convert each CAMPD emission from short tons to kilograms using 1 short ton = 907.18474 kilograms.
- Look up the TRACI factor using the pollutant, air medium, impact category, and TRACI version.
- Multiply the pollutant mass by its factor and sum pollutant contributions only within the same impact category.
- Optionally divide the category result by gross load to obtain an impact-intensity result per MWh.

The unit conversion is:

$$Mass_{i,kg} = Mass_{i,short\ ton} \times 907.18474$$

For impact category c, the total potential impact is calculated across the matched pollutants:

$$Impact_c = \Sigma_i (Mass_{i,kg} \times CF_{i,c})$$

Here, i identifies a pollutant and CF is its TRACI characterization factor for category c. The output unit comes from TRACI, such as kilograms of CO2 equivalent or kilograms of SO2 equivalent. Results from different impact categories must remain separate because their units and meanings are different.

The optional impact intensity is:

$$Impact\ intensity_c = \frac{Impact_c}{Gross\ load_{MWh}}$$

- Display selected potential-impact results by facility, unit, state, fuel, or year
- Show each impact category in its own equivalent unit
- Optionally divide the result by gross load to report impact per MWh
- Record the TRACI version and factor used in each calculation

The results represent potential environmental or health impacts for comparison; they do not measure actual illness, local exposure, or proven environmental damage.

5 Prediction Module

5.1 Prediction Objectives

Each team should select one clearly defined prediction target.

Possible regression targets include:

- Annual or monthly CO2, SO2, or NOx emissions
- Gross load or heat input
- Emission intensity
- Heat rate

Possible classification targets include:

- High- versus low-emission unit
- Whether emissions exceed a defined threshold
- Whether emission intensity belongs to a high-risk category
- Whether a facility's emissions will increase in the next period

5.2 Candidate Features

- Historical emissions
- Gross load, operating time, and heat input
- Fuel type, unit type, and state
- Pollution-control technologies
- Previous-year or previous-period values
- Calendar and facility characteristics

Students must prevent target leakage. A feature that directly reveals the target should not be used to claim genuine predictive performance.

5.3 Candidate Models

At least two models should be compared with a simple baseline.

Regression options:

- Mean or median baseline
- Linear, Ridge, or Lasso regression
- Decision-tree regression
- Etc.

Classification options:

- Majority-class baseline
- Logistic regression
- Decision tree
- Etc.

5.4 Model Evaluation

For multi-year data, a chronological split is generally more appropriate than a random split. Earlier years should be used for training, a later period for validation, and the most recent period for final testing.

Regression metrics may include:

- Mean absolute error
- Root mean squared error
- Coefficient of determination R squared
- Mean absolute percentage error when appropriate

Classification metrics may include:

- Accuracy, precision, recall, and F1 score

6 Sensitivity and Scenario Analysis

6.1 Purpose

Sensitivity analysis examines how a fitted model's output changes when selected input values change. For example, the user might increase gross load by 10 percent or reduce heat input by 5 percent and observe the change in predicted emissions. Sensitivity analysis describes model behavior; it does not by itself prove a real-world causal effect.

6.2 One Variable at a Time Analysis

The user should select:

- A trained model
- A baseline facility or unit
- One input variable
- A minimum and maximum value
- A step size or percentage change

The system should hold other variables constant and report the input value, predicted result, absolute change, percentage change, and a sensitivity curve.

6.3 Scenario Analysis

The system should support scenarios such as:

- Baseline values
- Gross load increased by 10 percent
- Heat input reduced by 5 percent
- Fuel or control-technology category changed
- A combined high-efficiency scenario

The application should compare each scenario with the baseline and warn users when a proposed value falls outside the model's training-data range.

6.4 Visualization of Sensitivity and Scenario Results

Sensitivity results should be displayed visually so users can see how predicted emissions or impact indicators respond to input changes. At minimum, the application should provide:

- A sensitivity curve with the varied input on the horizontal axis and the predicted result on the vertical axis
- A clearly marked baseline point and reference lines showing the baseline input and prediction
- A bar chart comparing the baseline prediction with each named scenario
- Labels or tooltips showing input values, predictions, absolute changes, and percentage changes
- A visual warning, such as a shaded region or different color, when an input is outside the model's training-data range

For each scenario, the percentage change from the baseline is:

$$Percentage\ change\ = \frac{Scenario\ prediction - Baseline\ prediction}{Baseline\ prediction} \times 100\%$$

The charts should use the same filters and units as the underlying analysis and should be available for image or data export. Users should interpret these graphics as model-response visualizations rather than evidence of a real-world causal effect.

7 Saving and Exporting Results

Users should be able to export:

- Filtered analytical datasets
- Summary-statistics and comparison results
- Model evaluation results and predictions
- Sensitivity and scenario-analysis results
- Chart images or chart data

Each export should include sufficient metadata for reproducibility:

- Analysis date and dataset identifier
- Selected filters and variables
- Definitions of derived indicators
- Model name, features, and parameters
- Training and testing periods
- Software or pipeline version

8 Proposed System Architecture

The Phase 2 processing flow is:

1. Acquire EPA CAMPD data through upload or retrieval.
2. Validate and store the data in the Phase 1 database.
3. Import the selected EPA TRACI characterization factors.
4. Build a filtered analytical dataset.
5. Use pandas and NumPy for transformation and descriptive analysis.
6. Calculate selected environmental and health impact indicators.
7. Create interactive visualizations.
8. Train and evaluate prediction models with scikit-learn.
9. Perform sensitivity and scenario analysis.
10. Present results through Flask pages or JSON endpoints.

8.1 Recommended Technologies

- Web application: Flask
- HTML rendering: Jinja2
- Page design: Bootstrap and custom CSS
- Browser interaction: JavaScript
- Interactive charts: Plotly.js
- Database: SQLite for development and PostgreSQL as an optional deployment upgrade
- Database interface: SQLAlchemy
- Data processing: pandas and NumPy
- Machine learning: scikit-learn
- Statistical analysis: SciPy or statsmodels when needed
- Impact assessment: EPA TRACI 2.2 characterization factors
- Model storage: joblib with model metadata stored in the database
- Testing: pytest and pytest-flask
- Version control: Git and GitHub

React is optional. It may be useful for dashboards with extensive client-side state and linked visualizations, but Flask, Jinja2, Bootstrap, JavaScript, and Plotly.js are sufficient for the required Phase 2 functions.

10 Development Milestones

1. Project definition and analytical data layer: finalize the analytical questions, prediction targets, data sources, and reporting years; implement multi-year CAMPD and TRACI import, missing-value handling, aggregation, and derived indicators.

2. Descriptive and impact analysis: implement summary statistics, grouping, comparisons, rankings, and the selected TRACI environmental and health impact calculations.

3. Visualization: create interactive emissions, operating, and impact charts with dashboard filters and export options.

4. Prediction: implement a baseline and at least two candidate models, separate training and testing data appropriately, and report evaluation results.

5. Sensitivity and scenario analysis: implement one-variable-at-a-time analysis, comparison scenarios, sensitivity curves, and scenario-comparison charts.

6. Integration and final delivery: connect the modules, test calculations and routes, improve usability, and complete the report, presentation, demonstration, and deployment.

The project schedule is six weeks, with one major milestone completed each week.

12 Required Deliverables

12.3 Source Code

- Clear GitHub commit history
- Installation and execution instructions
- Environment template and database migrations
- Automated tests and small test fixtures
- No API keys, confidential credentials, or unnecessarily large raw datasets

12.4 Project Report

- Refer to phase 1

12.5 Presentation and Demonstration

- Explain the analytical question
- Demonstrate the working application
- Interpret at least two visualizations
- Compare prediction models with a baseline
- Demonstrate sensitivity or scenario analysis
- Explain limitations